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CASE STUDY

Four teams. Three raw logs. Build the **single source of truth**

Fleet health and operational excellence, from fragmented operational logs to a governed analytical layer. Design the semantic model, the KPI framework and the dashboards that let Fleet, Ops, Collections and Finance act before value leaks — at 15,000 vehicles and beyond.

Role: Senior BI Engineer

Data Team · Confidential

15,000

Vehicles by EOY 2026
— the scale your
model must hold

3

Core data sources to
unify into one layer

4

Business areas served:
Fleet, Ops, Collections,
Finance

3

Dashboard audiences:
Leadership, Fleet &
Ops, Collections

› Business context

OneCarNow (OCN) is a fintech on a mission: to redefine mobility for gig economy drivers in Mexico. We are the market leader in lease-to-own solutions for Uber, DiDi and Rappi drivers, backed by **\$230M USD in funding**, and we are scaling fast — from **10,000 to 15,000 vehicles by EOY 2026**.

But operational visibility remains fragmented:

- › **Fleet** lacks real-time insight into inventory aging and readiness.
- › **Collections** cannot predict defaults early enough.
- › **Operations** struggles to align PARK logistics with delivery demand.
- › **Leadership** relies on manual reports instead of proactive dashboards.

As the Senior BI Engineer, you sit within the **Data Team** but serve as the trusted analytics partner to Fleet, Ops, Collections and Finance. Your mandate: **transform raw operational logs into a governed, trusted and actionable analytical layer**.

Daily

Decision cadence the solution must power

15K+

Vehicles the model must scale to — plus future markets

Semantic

You own the layer, KPIs, dashboard logic and alignment

Proactive

Idle alerts and risk scoring instead of manual reports

› Case objective

Design and deliver a **production-grade BI solution** that:

- › **Unifies data** from the three core sources listed in the next section.
- › **Powers daily decision-making** for Fleet and cross-functional teams.
- › **Enables proactive interventions** — e.g. idle vehicle alerts, default risk scoring.
- › **Scales** to 15K+ vehicles and future markets (e.g. Brazil).

Scope of ownership

You are **not responsible for building ML models or data pipelines** — but you own the **semantic layer, KPI definitions, dashboard logic and stakeholder alignment**.

› Data provided

SOURCE	DESCRIPTION	KEY FIELDS
db_stock_vehicles.csv	Master vehicle inventory.	vehicle_id , status , category , subCategory , brand , model , isElectric , deliveredDate , createdAt , updatedAt
db_stock_update_history.csv	Audit log of system events.	vehicle_id , update_type (e.g. "CONDUCTOR ASIGNADO", "GPS instalado"), timestamp , user_id
db_stock_vehicle_status_histories.csv	History of vehicle category and status changes.	vehicle_id , category , sub_category , previousCategory , previousSubCategory , createdAt , updatedAt

Note

You may assume these represent **daily snapshots from MongoDB and PostgreSQL**.

› Expected scope

- 1 Data understanding & quality assessment
- › **Profile data completeness:** are all vehicles represented across sources?
 - › **Identify gaps:** missing VINs, inconsistent status labels, duplicate records.
 - › **Propose cleaning rules:** standardize category / subCategory (e.g. "workshop" → "in_maintenance"); resolve conflicting timestamps (e.g. deliveredDate vs. the "CONDUCTOR ASIGNADO" event).
 - › **Flag risks:** vehicles with inconsistent timestamps across creation, update or status events.
- 2 Analytical data model design
- › Design a **semantic layer optimized for business consumption**.
 - › **Justify granularity:** event-level for diagnostics, daily snapshot for KPIs.

3 KPI framework — aligned to business needs

AREA	KPI	FORMULA	OWNER
Fleet Health	% Unproductive Fleet	(idle + workshop + withdrawn) / total fleet	Fleet Manager
Operational Velocity	Avg. Days to Ready	AVG(days from receipt to "driver_assigned")	Ops Lead
Collections Risk	Default Risk Score	% of vehicles > no payment in 7d	Collections Head
Asset Efficiency	Inventory Turnover	Vehicles assigned / Avg. monthly stock	CFO

Define **refresh frequency (daily)**, **aggregation logic** and **data freshness SLA**. Beyond the predefined KPIs, **propose additional KPIs or indicators** that can help answer relevant business questions or identify operational inefficiencies.

4 Dashboard design

Build a **multi-audience dashboard**:

Tab 1 · Fleet Health Overview

For Leadership

- › Fleet composition (active, idle, workshop, withdrawn).
- › Trend: % unproductive fleet (MoM).
- › Alert: top 50 vehicles >45 days in stock.

Tab 2 · Operational Deep Dive

For Fleet & Ops

- › PARK-level inventory heatmap.
- › Workshop turnaround by brand / model.
- › GPS installation vs. driver assignment gap.

Tab 3 · Risk & Recovery

For Collections

- › Default risk score distribution.
- › Correlation: status transitions vs. time in inventory.
- › Early warning list: vehicles with prolonged or unresolved status.

Justify every visualization — e.g. "this enables Fleet to reassign idle units before depreciation accelerates."

5 Stakeholder management & governance

- › **Requirements gathering**: how will you translate Fleet's "I need to reduce idle time" into a measurable KPI?
- › **KPI ownership**: who validates the definition of "ready for assignment"?
- › **Change control**: how are new event types (e.g. "VEHICULO REINGRESADO A STOCK") incorporated?
- › **Data dictionary**: document business definitions (e.g. "Unproductive = not earning + not in transit").

6 Implementation & adoption plan

Quick wins

- › Daily email: "Top 10 vehicles >30 days idle".
- › Slack alert: "PARK CDMX at 95% capacity".

Success metrics

- › $\geq 80\%$ of the Fleet team using the dashboard weekly.
- › $\leq 48h$ resolution time for data discrepancy requests.

› Deliverables

Submit:

1 · Executive summary

One to two pages: **problem, approach, impact.**

2 · Dashboard

Tableau, Power BI or Looker — **with annotated logic.**

3 · KPI & data dictionary

Definitions, formulas and owners for every metric you put on screen.

4 · Implementation plan

Timeline, risks and adoption strategy.

› Evaluation criteria

CRITERION	WHAT WE LOOK FOR
Business acumen	Do your KPIs solve real operational problems?
Data rigor	Is your model clean, consistent and auditable?
Stakeholder fluency	Can you bridge Data ↔ Fleet ↔ Collections?
Execution clarity	Is your plan realistic, phased and measurable?
Governance mindset	Are definitions controlled and documented?

"At OCN, your dashboards don't just show data — **they prevent defaults, accelerate deliveries and keep drivers earning.** You are the translator between bytes and business outcomes."

OneCarNow Data & Fleet Leadership